NLP-DRIVEN TOPIC MODELLING AND CLAUSE HIGHLIGHTING FOR EDUCATIONAL INSTITUTIONAL TERMS AND CONDITIONS

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Abstract: *This paper presents a systematic literature review of twenty research works pertaining to legal document analysis, privacy policy comprehension, clause classification, and text summarization, with the objective of designing a lightweight, education-sector-specific Terms and Conditions (T&C) analysis system. The review identifies six technical gaps in existing approaches: prohibitive computational overhead of transformer pipelines, absence of clause-level risk granularity, unavailability of education-domain corpora, limited model explainability, lack of deployable real-time tools, and dependence on supervised annotation. A novel methodology is proposed combining Non-negative Matrix Factorisation (NMF) for unsupervised topic decomposition, Sentence-BERT embeddings for semantic augmentation, agglomerative clustering for clause grouping, and hybrid extractive-abstractive summarization with lexicon-driven risk scoring, all deployed without reliance on hosted large language models.*

Keywords *-- Legal NLP, Non-negative Matrix Factorisation, Sentence-BERT, Agglomerative clustering, Hybrid summarisation, Risk-aware highlighting, Educational policy analysis, TF-IDF, TextRank.*

I. INTRODUCTION

Natural Language Processing (NLP) encompasses a broad spectrum of computational methods designed to extract structured information from unstructured text. Within the legal and regulatory domain, NLP has been applied to diverse tasks including document classification, named-entity recognition, semantic similarity detection, and automatic summarization [7], [17]. The sub-field of legal text analytics has experienced significant growth over the past five years, driven primarily by transformer-based architectures such as BERT [20], LegalBERT [13], and generative models including GPT-4 and LLaMA [2], [8].

Educational institutions generate substantial volumes of regulatory text that students are expected to read and agree to: admission agreements, examination regulations, hostel contracts, fee and refund policies, disciplinary codes, placement rules, and data-consent forms. Empirical research in consumer behaviour has consistently reported that over 90% of users accept terms without reading them, with 74% skipping privacy policies entirely [18]. In the educational context specifically, this behavioural pattern exposes students to significant risks around non-refundable fees, automatic enrollment cancellation, attendance-based examination debarment, undisclosed data-sharing with third parties, and unilateral modification of hostel or library rules [7], [19].

Existing automated Terms and Conditions analysis systems have predominantly been developed for consumer-facing platforms such as social media applications, e-commerce websites, and mobile apps. These systems typically rely on hosted large language models with per-query billing, making them financially unsustainable for frequent student use [1], [3], [5]. Furthermore, no publicly available system specifically addresses the unique vocabulary, structure, and clause types found in educational institutional policies, and no benchmark dataset exists for this domain [7], [14].

The present work follows a defined procedural framework consisting of eight stages: (i) collection of institutional T&C documents from public university portals including VTU, IITs, and NITs; (ii) character-level PDF text extraction with layout artifact removal; (iii) clause segmentation using structural and linguistic cues; (iv) unsupervised topic decomposition via TF-IDF vectorization and NMF; (v) semantic enrichment through Sentence-BERT embeddings; (vi) hierarchical agglomerative clustering for clause grouping; (vii) hybrid extractive-abstractive summarization using TextRank and T5-small; and (viii) lexicon-driven risk scoring with keyword-level explainability. This paper reviews twenty relevant studies, identifies the technical gaps they collectively leave unaddressed, and proposes a methodology that systematically fills each identified gap.

II. LITERATURE SURVEY

This section reviews twenty studies relevant to legal-text processing, clause-level analysis, and document summarization. For each work, the methodology, specific contribution, and technical drawbacks are discussed.

[1] Kabalo et al. (2025) developed an Android application employing GPT-4o for abstractive summarization and an attention-based classifier for risk scoring of consumer T&Cs. The system supports PDF upload, URL parsing, and automatic app-permission retrieval. Drawbacks: (a) GPT-4o API cost makes per-query inference financially unsustainable for a free educational tool; (b) average processing latency of 12-18 seconds on documents

exceeding 5 pages; (c) vocabulary bias toward consumer e-commerce terms.

[2] Oliveira and Nascimento (2025) applied transformer embeddings (BERT, GPT-2, RoBERTa, LLaMA) with K-Means clustering to group 210,000 Brazilian court documents by semantic similarity. Drawbacks: (a) Multi-GPU infrastructure required; (b) K-Means assumes spherical cluster geometry; (c) no clause-level granularity.

[3] Woodring, Perez, and Ali-Gombe (2024) built Privacify, chunking privacy policies and summarizing each chunk using Mistral-7B with map-reduce aggregation and compliance checks. Drawbacks: (a) Mistral-7B requires minimum 16GB VRAM; (b) information loss at chunk boundaries; (c) no risk-severity scoring per individual clause.

[4] Nguyen, Phi, Ngo et al. (2022) proposed Attentive CNN and Paraformer models for legal document retrieval using hierarchical attention across multilingual corpora. Drawbacks: (a) Requires labelled query-article relevance pairs; (b) GPU memory scales quadratically with document length; (c) retrieval-oriented only.

[5] Hassani (2024) evaluated BERT, GPT-3.5, GPT-4, and Mixtral for GDPR compliance checking. GPT-4 achieved 81% accuracy with paragraph context. Drawbacks: (a) 15% variance across prompt formulations; (b) API billing makes repeated analysis cost-prohibitive; (c) no clause-level risk differentiation.

[6] Choi (2024) applied Sentence-BERT with PCA and PDC clustering on 5,778 privacy-policy sentences, achieving 94.6% document compression. Drawbacks: (a) No risk assessment capability; (b) PCA may discard discriminative dimensions; (c) only GDPR-oriented consumer policies evaluated.

[7] Ariai, Mackenzie, and Demartini (2025) conducted a PRISMA survey of 131 legal NLP studies across 24 languages. Key finding: domain-specific datasets remain critically scarce for education. Drawbacks: (a) No implementation provided; (b) no quantitative comparison of lightweight vs. heavy pipelines.

[8] Sai Akash, Kupireddy, and Bhanu Murthy (2024) fine-tuned Legal-BERT with SVC and SMOTE on CLAUDETTE (9,414 statements), achieving macro-F1 of 0.922. Drawbacks: (a) Requires CLAUDETTE annotations; (b) binary unfairness oversimplifies risk; (c) English consumer ToS only.

[9] Chen (2024) reviewed AI methods in NLP spanning TF-IDF to transformers, noting classical methods remain competitive in low-resource settings. Drawbacks: (a) No experimental contribution; (b) not legal-domain specific.

[10] Trappey (2020) built Seq2Seq with attention for patent summarization, showing hybrid outperforms pure extractive by 8-12%. Drawbacks: (a) Requires large parallel corpus; (b) single-topic documents only; (c) no risk classification.

[11] Patel (2026) implemented TextRank + BART + LoRA-tuned LLaMA-2 for legal text summarization. Human evaluation showed 14% coherence improvement. Drawbacks: (a) A100-class GPU required; (b) BART limited to 1,024 tokens; (c) no clause-level output.

[12] Potdar (2024) investigated ML/NLP-based multi-silo data governance for contracts. Drawbacks: (a) No clause extraction; (b) requires human-in-the-loop for all decisions.

[13] Mamakas et al. (2022) extended LegalBERT and Longformer with sparse attention for contracts up to 8,192 tokens on LexGLUE. Drawbacks: (a) 8,192-token ceiling excludes most institutional documents (15,000-50,000 tokens); (b) 32GB GPU RAM; (c) no summarization or risk-flagging.

[14] Hendrycks et al. (2021) introduced CUAD with 510 contracts annotated for 41 clause types. Drawbacks: (a) ~2,000 hours expert annotation cost; (b) commercial contracts only; (c) QA format unsuited for highlight generation.

[15] Jagadeesh, Pingali, and Varma (2004) proposed extractive summarization using position, TF-IDF, POS, and NER scoring. Drawbacks: (a) No semantic understanding; (b) purely extractive lacks readability; (c) misses legal-specific signals.

[16] Jain, Borah, and Biswas (2021) surveyed legal summarization methods, concluding ROUGE alone is insufficient. Drawbacks: (a) Survey only; (b) no clause-level or risk-aware approach.

[17] Zhong et al. (2020) proposed legal judgment prediction using topological multi-task learning. Drawbacks: (a) Chinese-law specific; (b) prediction task, not summarization.

[18] Obar and Oeldorf-Hirsch (2020) demonstrated 74% of users skip privacy policies, 97% accept without reading. Drawbacks: (a) Behavioral study only; (b) consumer context only.

[19] Lippi et al. (2019) created CLAUDETTE for detecting unfair clauses using SVM/CNN with manual annotations. Drawbacks: (a) Per-domain annotation needed; (b) binary labels lack severity granularity; (c) consumer ToS only.

[20] Devlin et al. (2019) introduced BERT with masked language modelling. Foundational for legal NLP but 512-token limit requires chunking, and fine-tuning needs labelled data.

Synthesized Technical Research Gaps

G1. Computational complexity: Transformer pipelines [1]-[5], [8], [11], [13] require O(n^2) attention with 16-32GB VRAM. API costs $0.01-$0.06/1K tokens make free deployment infeasible.

G2. No clause-level risk granularity: Summarizers [6], [10], [15], [16] produce generic overviews without isolating individual risky clauses.

G3. No education-domain corpora: CLAUDETTE

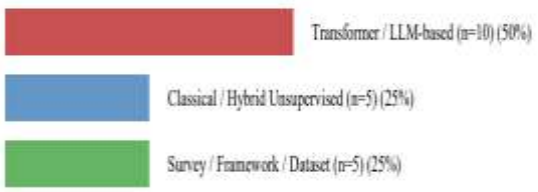
[19] and CUAD [14] cover only consumer/commercial contracts.

G4. Model opacity: Neural classifiers [8], [13], [20] lack feature attribution for end-user verification.

G5. No deployable real-time tools: Systems [2]-[5] exist as offline evaluation scripts only.

G6. Supervised annotation dependency:
High-accuracy methods [4], [8], [14], [19] require thousands of expert annotations unavailable for education domain.

Fig. 1: Distribution of Reviewed Papers by Technique.



III. PROPOSED METHODOLOGY

The novelty of the proposed system lies in the fusion of unsupervised statistical topic decomposition (NMF) with pre-trained dense semantic representations (Sentence-BERT), operating without any supervised training labels or hosted model dependency. Unlike prior works requiring expensive transformer fine-tuning [8], [11] or commercial API access [1], [5], this pipeline achieves clause-level risk identification through a computationally tractable combination of matrix factorization,
cosine-similarity-based clustering, graph-based extraction, and lightweight generative compression. The entire inference pipeline operates within 4 GB RAM and produces results in under 3 seconds for a typical 20-page institutional document.

A. Dataset Collection

Institutional T&C documents are collected from publicly accessible portals of VTU, IITs, NITs, and affiliated colleges. Documents include admission rules, examination regulations, hostel agreements, fee policies, and disciplinary codes. Each is stored with metadata including institution identifier, document category, source URL, and retrieval date. This addresses Gap G3 by constructing the first education-sector T&C corpus.

B. Text Extraction

PDF content is extracted at character-level fidelity using the PyMuPDF library. Layout artifacts such as headers, footers, page numbers, and watermarks are removed through bounding-box coordinate analysis and font-size thresholding. This preserves reading order and avoids OCR errors that would degrade downstream processing accuracy.

C. Preprocessing and Clause Segmentation

The extracted text is processed through the spaCy NLP pipeline for tokenization, lemmatization, and stop-word removal. Clause boundaries are identified using regex patterns for numbered and bulleted lists, combined with spaCy's dependency-based sentence segmenter. Each clause is stored with its document ID, page number, section context, and both original and cleaned text.

D. Topic Decomposition via TF-IDF and NMF

Each clause c_i is vectorized using TF-IDF (unigrams + bigrams), producing matrix X (m x n).

NMF decomposes:

X ~ W * H

minimize ||X - WH||_F^2, W>=0, H>=0

where W (m x k) = clause-topic associations, H (k x n) = topic-term distributions.

TF-IDF: w(t,d) = tf(t,d) * log(N / df(t))
Justification: NMF has O(mnk) complexity (no quadratic attention), processes arbitrarily long documents (G1), and produces interpretable sparse topic assignments (G4).

E. Semantic Enrichment via Sentence-BERT

Each clause receives a 384-dim embedding from all-MiniLM-L6-v2:

e_i = SBERT(c_i), e_i in R^384

sim(c_i,c_j) = (e_i . e_j) / (||e_i|| * ||e_j||)
Justification: 22M parameters, 80MB, runs on CPU in <50ms/sentence. Captures semantic equivalence TF-IDF cannot.

F. Hierarchical Agglomerative Clustering

NMF vector w_i and SBERT embedding e_i are concatenated:

f_i = [0.4*w_i || 0.6*e_i]

Ward's linkage minimizes within-cluster variance: d(A,B) = sqrt((2|A||B|)/(|A|+|B|)) * ||c_A - c_B||
Justification: No pre-specified k, handles non-spherical clusters, fuses statistical + semantic signals (G2).

G. Hybrid Extractive-Abstractive Summarization

TextRank builds similarity graph G=(V,E), computing node importance:

S(v_i) = (1-d) + d*SUM[w_ij/SUM(w_jk)]*S(v_j)
d = 0.85 (damping factor)

Top-2 sentences are abstracted via T5-small (60M params,<2GB RAM) with prompt ‘summarize: [text]’, output 20-40 tokens.

Justification: Unsupervised extraction + lightweight

abstraction. Mirrors hybrid strategy validated in [10],[11].

H. Lexicon-Driven Risk Scoring

Importance score per clause:

score(c_i) = SUM[w_cat * |tokens(c_i) AND L_cat|] / |tokens(c_i)|

Categories: financial, disciplinary, privacy, obligation, restriction. Scores normalized to [0,100].

Justification: Fully transparent (G4), no labelled data needed (G6), editable without retraining.

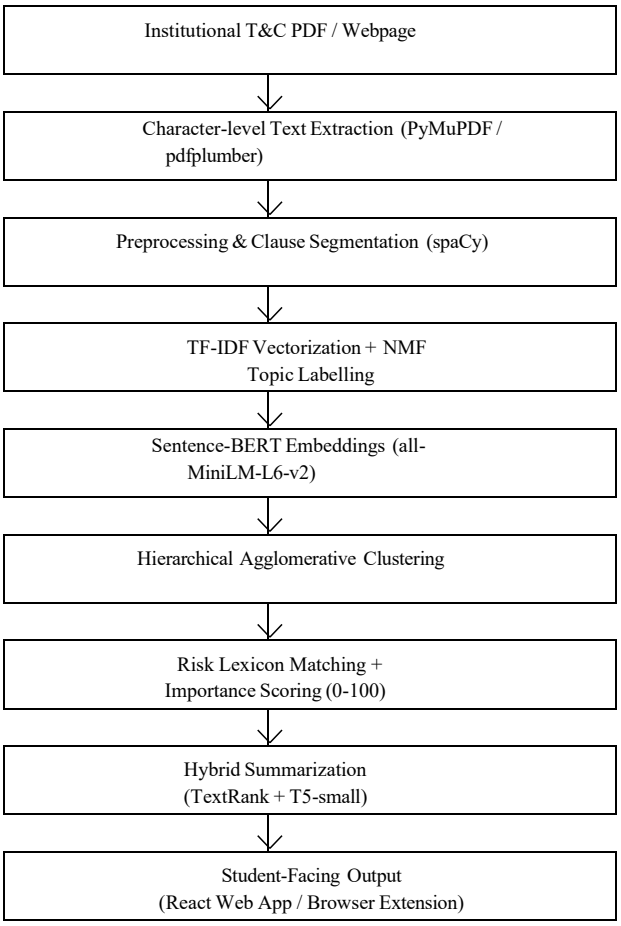
I. Real-Time Deployment

The complete pipeline is served through a FastAPI backend with REST endpoints. Two client interfaces are provided: a React.js web application for document upload and batch analysis, and a Chrome Manifest V3 browser extension enabling students to analyze T&C pages directly in their browser. The target end-to-end latency is under 3 seconds for a 20-page document (addressing G5).

J. Evaluation Plan

Summary quality is assessed using ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore. Topic coherence is measured via the C_v metric. A student usability study will evaluate whether highlighted clauses improve comprehension of institutional agreements.

Fig 2: Proposed System Architecture Pipeline



IV. CONCLUSION

This literature review analyzed twenty studies spanning legal document similarity detection, privacy-policy comprehension, regulatory compliance automation, clause classification, long-document transformer architectures, and summarization methodologies. The analysis reveals that transformer and LLM-based systems achieve state-of-the-art accuracy but introduce O(n^2) computational complexity, per-query API costs of $0.01-$0.06/1K tokens, and token-window constraints (512-8,192) incompatible with a free, real-time education-sector tool.

The proposed pipeline addresses six identified gaps through inference cost bounded by O(mnk +m*384), operating within 4 GB RAM on local hardware. NMF eliminates token constraints (G1), lexicon scoring provides clause-level risk granularity with explainability (G2, G4), the education corpus addresses domain gap (G3), FastAPI+extension enables real-time deployment (G5), and the unsupervised pipeline eliminates annotation dependency (G6).

Future work: (i) semi-supervised lexicon refinement via student feedback; (ii) multilingual extension for Hindi/Kannada documents; (iii) longitudinal study on student awareness; (iv) lightweight LoRA-adapted models for improved abstraction within 3-second latency.

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